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Proterorhinus semilunaris

(Freshwater Tubenose Goby)

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Piet Spaans

Proterorhinus semilunaris (Heckel, 1837)

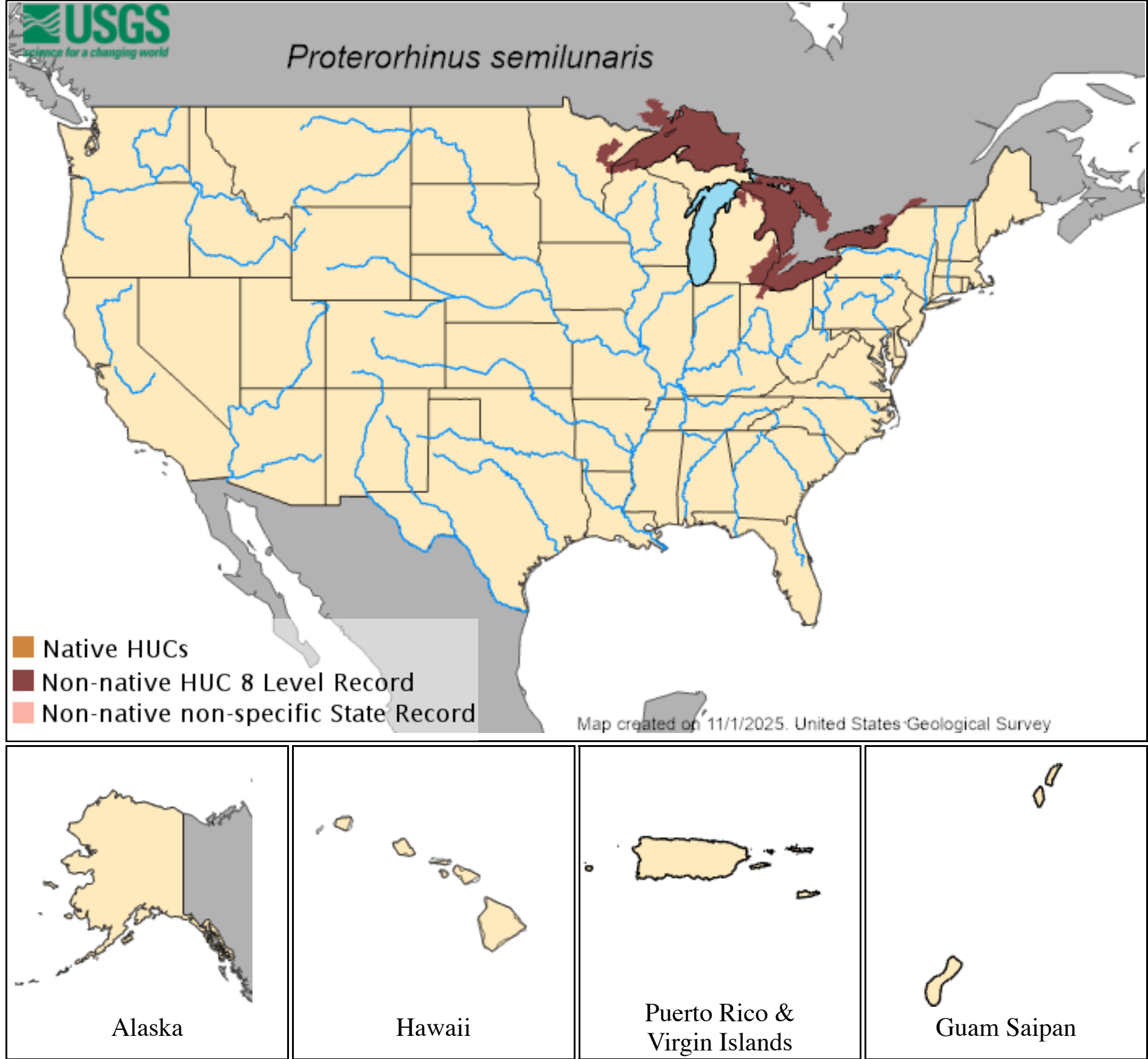
Common name: Freshwater Tubenose Goby

Synonyms and Other Names: *P. marmoratus* of authors, not of Pallas 1814; See Stepien and Tumeo 2006 for name change.

Identification: Characteristics were given by Berg (1949) and Miller (1986). This species (along with the round goby *Neogobius melanostomus*) can be distinguished from all other fishes in the Great Lakes by the presence of fused pelvic fins. Tubenose goby can be distinguished from the round goby by its long anterior nostrils and lack of black spot on posterior base of dorsal fin (Miller 1986; Jude 1993). Miller (1986), Crossman et al. (1992), and Jude et al. (1992) provided characteristics to distinguish the round and tubenose gobies.

Size: 12.7 cm total length

Native Range: Slightly brackish to freshwater. Eurasia, primarily in rivers and estuaries of the Black Sea basin; also in rivers of northern Aegean (Miller 1986, Freyhof and Naseka 2007; Neilson and Stepien 2009).



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: [Point Distribution Maps](#)

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Proterorhinus semilunaris* are found [here](#).

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
MI	1990	2024	11	Cheboygan ; Clinton ; Detroit ; Huron ; Lake Erie ; Lake Huron ; Lake St. Clair ; Lone Lake-Ocqueoc ; Ottawa-Stony ; Raisin ; St. Clair
MN	2001	2022	2	Lake Superior ; St. Louis
NY	2016	2024	2	Headwaters St. Lawrence River ; Raisin River-St. Lawrence River
OH	2001	2019	2	Cedar-Portage ; Lake Erie
PA	2011	2022	1	Lake Erie
WI	2003	2025	2	Beartrap-Nemadji ; St. Louis

Table last updated 12/3/2025

† Populations may not be currently present.

Ecology: The tubenose goby is a benthic omnivore, consuming a wide variety of benthic invertebrates (chironomids, crustaceans, copepods, dipterans, ephemeropterans, ostracods, trichopterans, gammarids, isopods and water bugs) and occasionally larval fishes (French and Jude 2001; Adamek et al. 2007).

Tubenose gobies generally inhabits shallow (less than 5 m depth), slow-moving, nearshore environments. They prefer areas with abundant aquatic macrophytes, but can also be found in sandy areas (Jude and Deboe 1996, Eros et al 2015). Eggs are typically attached to vegetation (Cavender, personal communication).

Means of Introduction: Introduced via ballast water.

Status: This species is reproducing and overwintering in the St. Clair River, and in Lake St. Clair, Michigan (Jude 1993; Cavender, personal communication). Beginning in 2016, tubenose goby has been consistently captured in annual surveys in the Thousand Island area of the upper St. Lawrence River (Goretzke et al. 2019). However, it remains relatively rare in even these environments even 30 years after first introduced. This species is not spreading rapidly (Vanderploeg et al. 2002), but has undergone discontinuous expansion such that it is found in localized nearshore areas of 4 of the 5 Great Lakes (Vanderploeg et al 2002; Kocovsky et al. 2011; Grant et al. 2012; MISIN 2019; Bowen 2018). Active adult movement and passive downstream larval drift could serve as a secondary invasion pathway along the St. Lawrence Seaway (Goretzke et al. 2019).

Tubenose goby is predicted by the GARP model to become established in the shoreline areas of all five Great Lakes (U.S. EPA 2008). Their distribution around the inshore areas of the Black and Caspian Seas indicates their potential for widespread occupation of inshore habitats where cover, especially plants, occurs in the lower Great Lakes (Jude et al 1992). Presence of round goby (*Neogobius melanostomus*) which expanded more rapidly in the Great Lakes may inhibit the population size of tubenose goby.

Impact of Introduction: The tubenose goby does not feed on zebra mussels, as do round gobies (Vanderploeg et al. 2002). However, it has been shown to have a significant overlap in diet preference with rainbow darters (*Etheostoma caeruleum*) and northern madtoms (*Noturus stigmosus*) and may compete with these native fish for food (French and Jude 2001).

Remarks: All tubenose gobies were previously included in a single species, *P. marmoratus*. Recently, *P. marmoratus* was restricted to marine/brackish populations in the Black Sea, and several names were resurrected/created for freshwater populations of tubenose gobies in different regions: *P. nasalis* and *P. semipellucidus* for populations inhabiting the Caspian Sea and Volga River basins (Freyhof and Naseka 2007; Neilson and Stepien 2009); *P. tataricus* endemic to several rivers on the Crimean Peninsula, Ukraine (Freyhof and Naseka 2007); and *P. semilunaris* for tubenose gobies in rivers and estuaries in the Black, Azov, and Aegean Sea basins (Freyhof and Naseka 2007; Neilson and Stepien 2009). *Proterorhinus semilunaris* is the only species of tubenose goby that has been introduced to North America (Stepien and Tumeo 2006; Neilson and Stepien 2009), and has also been introduced into several areas of central and western Europe (e.g., Manné and Poulet 2008; Cammaerts et al. 2011).

Although *P. semilunaris* is widely dispersed among drainages within the Black Sea basin, it is threatened in certain locale. The tubenose goby is considered endangered in Greece in the Ayannis spring near the town of Seres due to pollution and human-induced habitat change (Economidis 1995). In the Greek State, the tubenose goby is protected by law No. 67/1981 (Economidis 1995). This goby may live as long as five years (Jude 1993).

Pettitt-Wade et al. (2015) examined trophic niche breadth, plasticity, and overlap between round and tubenose gobies in Lakes Superior and St. Clair using stable isotope analysis. They found a higher isotopic trophic position and generally higher isotopic niche breadth and plasticity in round gobies, with little overlap between size-matched round and tubenose gobies, and suggested that this increased isotopic niche breadth and plasticity has assisted in the establishment success of round goby in the Great Lakes (widely abundant and distributed vs. low abundance and localized distribution of tubenose goby).

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Other Resources:
[Great Lakes Waterlife](#)
[US Fish and Wildlife Service Ecological Risk Screening Summary for Proterorhinus semilunaris](#)
[FishBase Summary](#)

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